LAB 11: Partial Differential Equation

Objective: To numerically obtain the governing PDE of any physical system and solve it numerically.

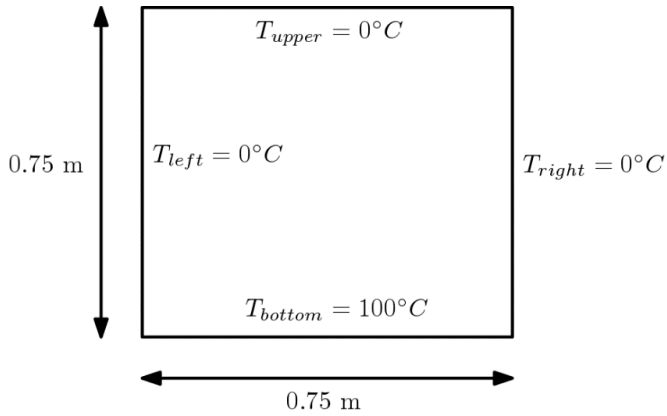
Theory:

Theory - Consider a 2D metal plate with dimensions 0.75x0.75 m2 is kept at 100 degree C at one of it end and rest of the ends are initially at 0 degree C as shown in the figure.

The transient Fourier law of heat conduction for 2D metal plate is given by,

$$\frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2}\right)$$

Where T(x,y,t) is the temperature at any location and time and x, y, t are space and time coordinates. Assume the thermal conductivity of plate is K = 1 W/m degree C.



Take the spatial discretization as delta x = delta y = 0.025 m.

Using explicit finite difference method, the discrete equation can be obtained using Taylor series expansion as below,

$$\frac{T_{i,j}^{n+1} - T_{i,j}^{n}}{\Delta t} = k\left(\frac{T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n}}{\Delta x^2} + \frac{T_{i,j+1}^{n} - 2T_{i,j}^{n} + T_{i,j-1}^{n}}{\Delta y^2}\right)$$

$$\text{Suppose } \Delta x = \Delta y = \Delta h$$

$$T_{i,j}^{n+1} = T_{i,j}^{n} + \frac{k\Delta t}{\Delta h^2}(T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n} + T_{i,j+1}^{n} - 2T_{i,j}^{n} + T_{i,j-1}^{n})$$

Take the spatial discretization as $\Delta x = \Delta y = \Delta h$ = 0.025 m and $\Delta t = 0.00015$, thermal conductivity of plate is K = 1 W/m degree C.

MATLAB program-

```
% Explicit method for 2D heat equation in flat plate
close all; clear all; clc;
L =0.75;
dx = 0.05;
```

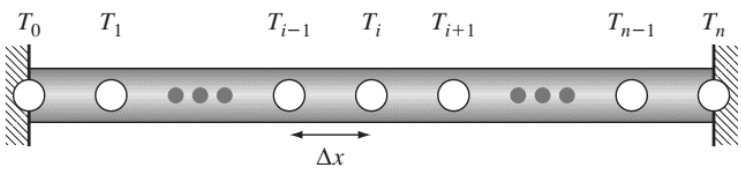
```
dy=0.05;
N = L/dx +1;
x = linspace(0,L,N);
y = linspace(0,L,N);
dt = 0.00015;
epsilon = 1e-4;
T_new = zeros(N,N);
 %boundary condition
      T_new(1,:) =100;
      T_new(:,1) =0;
      T_new(N,:)=0;
      T_new(:,N)=0;

error =1;iter=0;
while(error > epsilon)
    iter=iter+1;
      T = T_new;% update T every iteration
   for i=2:N-1
      for j =2:N-1
         T_new(i,j) = dt*((T(i+1,j)-2*T(i,j)+T(i-1,j))/dx^2 +(T(i,j+1)-2*T(i,j)+T(i,j-1))/dy^2) +
T(i,j);

      end
   end
   error = max(max(abs(T-T_new)));

figure(2);
hold on;
pause(0.001);
%     contourf(T_new);
    shading flat;colorbar;
    contourf(x,y,T_new,'ShowText','on')
colorbar;
xlabel('x meter');ylabel('y meter')

end
```

Assignment -

One numerical as suggested by Faculty.

Sample numerical.

Consider 1D heat conduction in metal rod as below.



1D heat conduction governing equation is given below,

$$\frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial x^2}\right)$$

Write down the discretized equation is explicit finite difference method and obtain the Temperature profile T(x,t). Take length of rod is 0.75 m, Δx= 0.025 m, $\Delta t = 0.00015$, thermal conductivity of plate is K = 1 W/m degree C.

Handwritten Notes

Exp -11

Taylor expansion in x dir.

Forward:

$$T_{i+1} = T_i + \Delta x \frac{\partial T}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 T}{\partial x^2} + \cdots$$

Backward:

$$T_{i-1} = T_i - \Delta x \frac{\partial T}{\partial x} + \frac{\Delta x^2}{2!} \frac{\partial^2 T}{\partial x^2} + \cdots$$

Summing,

$$T_{i+1} + T_{i-1} = 2T_i + 2 \frac{\Delta x^2}{2!} \frac{\partial^2 T}{\partial x^2} + \cdots$$

$$\therefore \frac{\partial^2 T}{\partial x^2} \approx \frac{T_{i+1} - 2T_i + T_{i-1}}{\Delta x^2}$$

Similarly for $y \rightarrow$

$$\frac{\partial^2 T}{\partial y^2} \approx \frac{T_{i,j+1} - 2T_{i,j} + T_{i,j-1}}{\Delta y^2}$$

Now Time,

$$T_{i,j}^{n+1} = T_{i,j}^{n} + \Delta T_{i,j}^{n} \frac{\partial T}{\partial t} + \cancel{\cdots}$$

$$\therefore \frac{\partial T}{\partial t} \approx \frac{T_{i,j}^{n+1} - T_{i,j}^{n}}{\Delta t}$$

$$\therefore \frac{T_{i,j}^{n+1} - T_{i,j}^{n}}{\Delta t} = k \left(\frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} \right)$$

$$\Rightarrow \quad = k \left(\frac{T_{i+1,j}^{n} - 2T_{i,j}^{n} + T_{i-1,j}^{n}}{\Delta x^2} + \left(\frac{\partial^2 T}{\partial y^2} \right) \right)$$

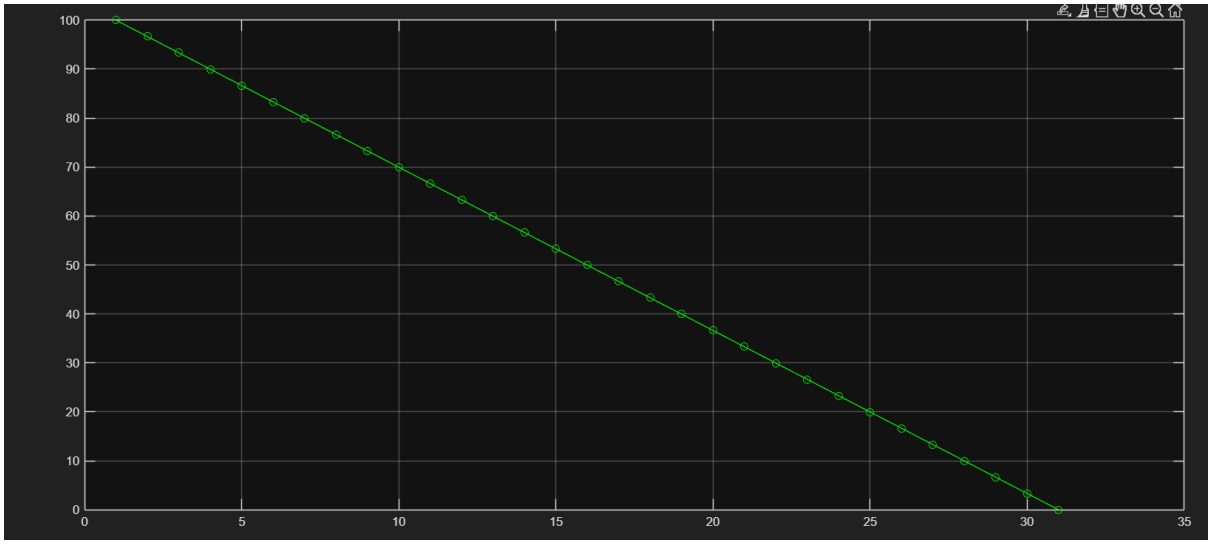
Sol

For 1D rod, $\frac{\partial T}{\partial t} = k\left(\frac{\partial^2 T}{\partial x^2}\right)$

$$\therefore \frac{T^{m+1} - T^m}{\Delta t} = k\left[\frac{T_{p+1}^m - 2T_p^m + T_{p-1}^m}{\Delta x^2}\right]$$

$$\Rightarrow T_p^{m+1} = T_p^m + \frac{k\Delta t}{\Delta x^2}\left[T_{p+1}^m - 2T_p^m + T_{p-1}^m\right]$$

Code and Output

```
clc
clear
close all

%analyticalsolution
t_exact = 0:0.01:2;
y_exact = exp((20/23)*t_exact.^2.3 + 2.1*t_exact);

% h=0.2
h = 0.2;
t1 = 0:h:2;

y_e1 = zeros(size(t1));
y_m1 = zeros(size(t1));

y_e1(1) = 1;
y_m1(1) = 1;

for i = 1:length(t1)-1

% Euler
    y_e1(i+1) = y_e1(i) + h*y_e1(i)*(2*t1(i)^1.3 + 2.1);

% Mod Euler
    k1 = y_m1(i)*(2*t1(i)^1.3 + 2.1);
    y_pred = y_m1(i) + h*k1;
    k2 = y_pred*(2*(t1(i)+h)^1.3 + 2.1);

    y_m1(i+1) = y_m1(i) + (h/2)*(k1 + k2);
end

%h = 0.1
h = 0.1;
t2 = 0:h:2;

y_e2 = zeros(size(t2));
y_m2 = zeros(size(t2));

y_e2(1) = 1;
y_m2(1) = 1;

for i = 1:length(t2)-1

% Euler
    y_e2(i+1) = y_e2(i) + h*y_e2(i)*(2*t2(i)^1.3 + 2.1);

% Mod Euler
    k1 = y_m2(i)*(2*t2(i)^1.3 + 2.1);
    y_pred = y_m2(i) + h*k1;
    k2 = y_pred*(2*(t2(i)+h)^1.3 + 2.1);

    y_m2(i+1) = y_m2(i) + (h/2)*(k1 + k2);
end


figure
hold on


plot(t_exact, y_exact, 'c', 'LineWidth', 3)


plot(t1, y_e1, 'r-o', 'LineWidth',1.8, 'MarkerSize', 6)
plot(t1, y_m1, 'm-s', 'LineWidth', 1.8, 'MarkerSize', 6)


plot(t2, y_e2, 'b-o', 'LineWidth', 1.8, 'MarkerSize', 6)
```

```
plot(t2, y_m2, 'g-s', 'LineWidth', 1.8, 'MarkerSize', 6)

grid on
xlabel('t','FontSize',12)
ylabel('y','FontSize',12)
title('Euler vs Modified Euler','FontSize',14)

legend('Analytical (Exact)', ...
       'Euler h=0.2','Modified Euler h=0.2', ...
       'Euler h=0.1','Modified Euler h=0.1', ...
       'Location','northwest')

```